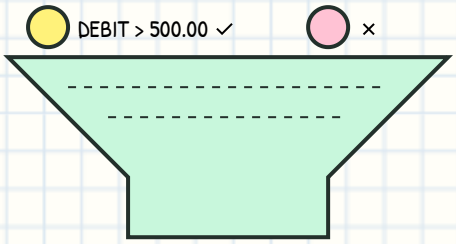
Filter · Map · Slice · my notes L07

the everyday intermediate operations, with the shared dataset

filter = a sieve · map = a stamping machine



60 txns → 11 pass (compareTo!)



1 in → 1 out · count stays 10

💰 the BigDecimal trap

`500.0 .equals(500.00)` → false

`500.0 .compareTo(500.00)` → 0

equals checks the **scale** too. Money → compareTo / signum.

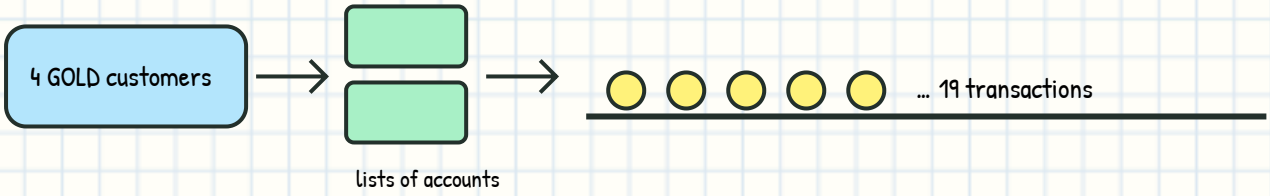
numbers? go primitive

`mapToInt(booked).sum()` → 1808

`mapToDouble(load).average()` → 84.1%

`.boxed()` → back to `Integer`

flatMap = open the boxes and pour them onto one belt 📦 → ●●●



map → Stream<List<Account>> (boxes still closed) · flatMap → Stream<Account>

mapMulti [16] = "call me for each result"

`(f, down) -> { if (≥ 95% full) down.accept(f.code()); }` → [SQ318, TR808, SQ21] · no Stream per element · write `.<String>mapMulti`

Practice set ✎

- debits > 500.00 → 11
- names uppercase → ALICE TAN …
- GOLD transactions → 19 (flatMap)
- distinct cities → 5
- first 5 by date → T001 … T005
- page 2, size 5 → T006 … T010

distinct = the bouncer checks equals

cities → **Singapore**, Kuala Lumpur, Jakarta, Colombo, Bangkok
(keeps the first one seen)

record CityTier → 9 of 10 ✓
plain class, no equals → 10 of 10 ×
1.0 vs 1.00 → 2 · setScale(2) → 1

peek = a window, not a worker

`list.stream().peek(save).count()`
count = 12, peeked = 0 (Java 9+)
debug only · never save / send / count

Slicing ✂ position vs condition

skip(5).limit(5) = page 2



skip 5 (T001–T005) T006 – T010

… page 13 → [] (no exception)

takeWhile(< 300) on UNSORTED payments

P001 120 | P002 45 | P003 310 ✋ | P004 …

→ [P001, P002] only
filter → 8 · sort first → 8 ✓

Worked example: GOLD spending review

filter GOLD → flatMap accounts → flatMap txns → filter DEBIT > 100 → sorted newest → skip/limit → 11 matches = pages of 5, 5, 1 · page 1 starts T060 194.28

Which operation? 🤔

- keep some → **filter** · keep a prefix of sorted data → **takeWhile**
- change each → **map** (numbers: **mapToInt**) · each becomes many → **flatMap** / **mapMulti**
- no duplicates → **distinct** · a page → **skip** + **limit** · just looking → **peek**

★ Remember

compareTo for money · map 1→1 · flatMap 1→many · mapMulti pushes · distinct needs equals · peek only to look · sort before limit/skip/takeWhile · next → L08 Sorting with Comparators